

WASP-39b

R-2089

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_120513 · created 2026-08-10T00:45:56+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456060.7498 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.011
Transit depth (Rp/Rs)^2	2.94 +/- 0.39 [%]
Orbital Inclination (inc)	87.25 +/- 0.42
Airmass coefficient 1 (a1)	1.198 +/- 0.012
Airmass coefficient 2 (a2)	-0.136 +/- 0.042
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	None
Optimal Aperture	3.68
Optimal Annulus	13.19
Transit Duration (day)	0.1136 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.011 vs 0.1457 (deviation 2.11σ)
Duration fit vs published	0.1136 d vs 0.12 d (deviation 1.47σ)
Mid-time O–C	-6.3 min
Depth SNR	14.3
Residual scatter	1.02 %

Light curve

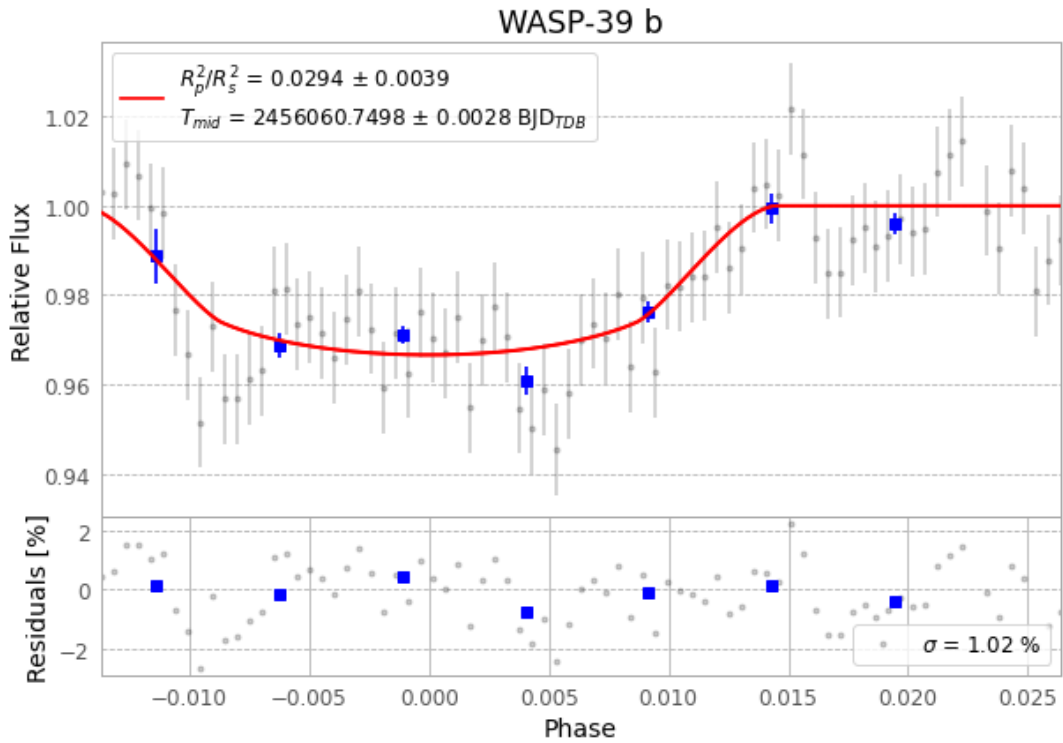


Figure 1 — FinalLightCurve_WASP-39 b_2012-05-12.png (SHA-256 26651ab32217e810...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d9a7a5124098182d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

79 FITS frames (52 MB), WASP-39120513043012.FITS → WASP-39120513082413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-120513.log (06cbb44287a3...), FinalLightCurve_WASP-39 b_2012-05-12.png (26651ab32217...), FinalLightCurve_WASP-39 b_2012-05-12.csv (296923c60b87...)

Compute resources

Wall time 141.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:45Z.